Can Machine Unlearning Remove Backdoors in Convolutional Neural Networks? Effects of Model Depth, Attack Stealth, and Partial Poison Identification

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I. Introduction

Modern AI models are increasingly trained on large-scale datasets collected by crawling the web, and while this has allowed monumental progress in machine learning, it is also impossible to verify the truth of each and every training example. As a result, incorrect information can be inserted into training data and then learned by models. Carlini et al. (2024) have shown that web-scale datasets are vulnerable to poisoning attacks [1]. In addition, training on user data raises serious privacy concerns, especially when a user wants to revoke access and wishes for their data to be deleted; however, their data may have already been used to train models, which are vulnerable to privacy attacks [2]. Together, these issues of data integrity and privacy show that machine unlearning is integral for the field to move forward.

II. Research Problem

How effectively can machine unlearning methods remove backdoor poisoning from CNNs, and how do model depth, attack stealth, and incomplete identification of poisoned data affect their success and accuracy cost?

III. Methods

A. Models

ResNet-18 and **ResNet-34** [3] are two convolutional neural networks (CNNs) built from residual blocks, which add skip connections to mitigate the degradation problem common in deeper networks.

Both models share the same architecture and differ mainly in depth: ResNet-18 has 18 layers and about 11 million parameters, while ResNet-34 has 34 layers and about 21 million parameters. This lets us study the effect of model depth on data poisoning and machine unlearning without architectural differences.

B. Dataset

CIFAR-10 and **CIFAR-100** [4] are two datasets that each contain 60,000 32×32 color images, split into 50,000 training and 10,000 test images. CIFAR-10 has 10 classes with 6,000 images per class, while CIFAR-100 has 100 classes with 600 images per class. Using both datasets allows us to study the effect of finer-grained classification on poisoned models and machine unlearning.

C. Data Poisoning Methods

We will evaluate three data poisoning methods of increasing stealth. Each method adds a trigger to a subset of training images and relabels them away from their ground-truth labels. The poisoned model will misclassify any input containing the trigger while behaving normally on clean inputs.

BadNets [5] poisons training images by stamping a small pixel pattern onto them as the trigger. The trigger is visible to humans, making BadNets a simple baseline.

FTrojan [6] embeds the trigger in the frequency domain. It transforms each image with a discrete cosine transform and perturbs selected frequency components, producing a trigger that is invisible to humans.

Adap-Blend [7] is an adaptive poisoning method. It hides the trigger by dividing it into a 4×4 grid and applying only 50% of the pieces to each poisoned image. At test time, the full trigger is used to activate the backdoor. Adap-Blend also adds regularization by keeping the correct labels on a portion of the triggered images, which weakens the link between the trigger and the backdoor during training and makes the poisoned samples harder to detect.

D. Machine Unlearning Methods

After a model has been poisoned, we will apply five methods to remove the backdoor. Each method receives an identified poison subset, made up of images that have been identified as poisoned, and a retained subset, made up of the remaining training data.

Retraining from scratch trains a new model on the retained subset only. It is the simplest but most expensive method and serves as our baseline. However, under partial identification, the unidentified poisoned images remain in the retained subset and can reintroduce the backdoor.

Fine-tuning continues to train the poisoned model on the retained subset, aiming to overwrite the backdoor.

SCRUB [8] is a teacher–student method in which the original poisoned model is the teacher and the student is initialized as a copy of the teacher. The student is trained to diverge from the teacher on the identified poison subset while converging to the teacher on the retained subset.

SalUn [9] is a saliency-based unlearning method. It computes a weight saliency map from the gradient of the loss on the identified poison subset and then updates only the most salient weights, removing the backdoor behavior while preserving performance on the retained subset.

Potion [10] is a training-free method. It identifies weights that are disproportionately important to the identified poison subset compared to the full training data and dampens them. Because no training occurs, the unidentified poisoned images in the retained subset cannot reintroduce the backdoor during unlearning.

E. Partial Identification

In practice, it is unrealistic to expect every poisoned image to be identified and removed. To reflect this, each unlearning method is given an identified poison subset, containing 25%, 50%, 75%, or 100% of the poisoned images, along with a retained subset made up of the remaining training data.

F. Evaluation

Attack Success Rate (ASR) is the percentage of triggered test images that the model misclassifies, excluding poisoned images that retain the ground truth. A successful unlearning method should reduce the ASR to near the level of a clean model.

Clean Accuracy (CA) is the model's accuracy on clean images. CA is used to measure the accuracy cost of each unlearning method. A successful unlearning method should maintain CA.

IV. Timeline

Week	Tasks
1	Experiment Setup • Set up environment. • Build data loaders for CIFAR-10/100. • Download all datasets, poisoning methods, and machine unlearning methods.
2	Clean Baseline Training • Train clean ResNet-18 and ResNet-34 on CIFAR-10/100.
3	Poison Datasets • Generate poisoned datasets with BadNets, FTrojan, and AdapBlend based on CIFAR-10/100. • Train ResNet-18 and ResNet-34 on the six poisoned datasets. • Verify attack success rate of each poisoned model. • Retrain ResNet-18 and ResNet-34 on retained subsets.
4	Unlearning Pipeline • Build an unlearning and evaluation pipeline. • Fine-tune poisoned models under full identification.
5	Full-Identification Experiments • Run SCRUB, SalUn, and Potion with all poisoned images identified.
6	Partial-Identification Experiments • Run experiments with all unlearning methods, with 25%, 50%, 75%, and 100% of poisoned images identified.
7	Analysis and Visualization • Analyze results across model size, attack, and identification level. • Create figures with the results. • Rerun experiments if needed.
8	Finalization • Write final report. • Create presentation.

Disclaimer

This proposal was prepared with the assistance of AI tools for citing references, grammar corrections, and LaTeX syntax.

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